

# Group & Ring Theory MATP33: Course Summary of Important Results

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## Lecture 0 — Background Results

**Theorem 1** (First Isomorphism Theorem). *Let  $f : G \rightarrow H$  be a group homomorphism. Then  $\ker f \trianglelefteq G$  and  $\operatorname{im} f \cong G/\ker f$ .*

**Theorem 2** (Second Isomorphism Theorem). *Let  $G$  be a group,  $H \leq G$ , and  $N \trianglelefteq G$ . Then  $HN \leq G$ ,  $H \cap N \trianglelefteq H$ , and  $HN/N \cong H/(H \cap N)$ .*

**Theorem 3** (Third Isomorphism Theorem). *If  $K \trianglelefteq G$  and  $K \leq L \trianglelefteq G$ , then  $(G/K)/(L/K) \cong G/L$ .*

**Theorem 4** (Correspondence Theorem). *Let  $N \trianglelefteq G$ . Every subgroup of  $G/N$  is of the form  $H/N$  where  $N \leq H \leq G$ , and conversely if  $N \leq H \leq G$  then  $H/N \leq G/N$ .*

**Theorem 5** (Lagrange). *If  $H \leq G$  and  $G$  is finite, then  $|H|$  divides  $|G|$ .*

**Theorem 6** (Fundamental Theorem of Finitely Generated Abelian Groups). *Let  $A$  be a finitely generated abelian group. Then*

$$A \cong \mathbb{Z}^s \oplus \mathbb{Z}/a_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/a_r\mathbb{Z},$$

where  $s \geq 0$  and  $a_1 \mid a_2 \mid \cdots \mid a_r$  are non-zero non-units in  $\mathbb{Z}$ . This decomposition is unique.

## Lecture 1

**Theorem 7** (1.10). *Let  $G$  be a group and  $X$  a set. Then:*

- (i) *If  $X$  is a  $G$ -set, then the action of  $G$  on  $X$  induces a homomorphism  $\varphi : G \rightarrow S_X$ ;*
- (ii) *Any homomorphism  $\varphi : G \rightarrow S_X$  induces an action of  $G$  on  $X$ .*

*Proof.* (i) Define  $\varphi : G \rightarrow S_X$  by  $(\varphi(a))(x) = a \cdot x$  for  $a \in G$  and  $x \in X$ . We must first check that  $\varphi(a) \in S_X$  for each  $a \in G$ , i.e. that the function  $x \mapsto a \cdot x$  is a bijection  $X \rightarrow X$ .

- *Injective:* Suppose  $a \cdot x = a \cdot y$ . Then  $x = a^{-1} \cdot (a \cdot x) = a^{-1} \cdot (a \cdot y) = y$ .
- *Surjective:* For any  $x \in X$ , we have  $a \cdot (a^{-1} \cdot x) = x$ , so  $x \in \operatorname{im}(\varphi(a))$ .

Hence  $\varphi(a) \in S_X$ . To see  $\varphi$  is a homomorphism, let  $a, b \in G$ . Then for all  $x \in X$ ,

$$(\varphi(ab))(x) = (ab) \cdot x = a \cdot (b \cdot x) = (\varphi(a))((\varphi(b))(x)) = (\varphi(a)\varphi(b))(x).$$

(ii) Define  $a \cdot x := (\varphi(a))(x)$ . Then  $(ab) \cdot x = (\varphi(ab))(x) = (\varphi(a)\varphi(b))(x) = a \cdot (b \cdot x)$ , and  $e \cdot x = (\varphi(e))(x) = x$ . Hence  $X$  is a  $G$ -set.  $\square$

**Theorem 8** (1.11 — Cayley's Theorem). *Every group  $G$  is isomorphic to a subgroup of the symmetric group  $S_G$ .*

*Proof.* By Theorem 1.10(i), left translation  $a \cdot x := ax$  gives a homomorphism  $\varphi : G \rightarrow S_G$ . If  $a \in \ker \varphi$ , then  $\varphi(a) = \text{id}_G$ , so  $ax = x$  for all  $x \in G$ , giving  $a = e$ . Hence  $\ker \varphi = \{e\}$ , and by the First Isomorphism Theorem  $G \cong G/\ker \varphi \cong \text{im } \varphi \leq S_G$ .  $\square$

**Definition 1** (Normal core). *Let  $G$  be a group and  $H \leq G$ . The normal core of  $H$  in  $G$  is*

$$\text{core}(H) := \bigcap_{x \in G} xHx^{-1}.$$

*It is the largest normal subgroup of  $G$  contained in  $H$ .*

**Theorem 9** (1.12). *Let  $G$  be a group and  $H \leq G$  a subgroup of index  $n$ . Then there is a homomorphism  $\varphi : G \rightarrow S_n$  such that  $\ker \varphi = \text{core}(H)$ .*

*Proof.* Since  $|G/H| = n$ , we have  $S_{G/H} \cong S_n$ . By Theorem 1.10(i), the action of  $G$  on the coset space  $G/H$  by left multiplication gives a homomorphism  $\varphi : G \rightarrow S_n$  with  $\varphi(a)(xH) = axH$ . Then

$$\begin{aligned} \ker \varphi &= \{a \in G \mid axH = xH \text{ for all } x \in G\} \\ &= \{a \in G \mid x^{-1}ax \in H \text{ for all } x \in G\} \\ &= \{a \in G \mid a \in xHx^{-1} \text{ for all } x \in G\} \\ &= \bigcap_{x \in G} xHx^{-1} = \text{core}(H). \end{aligned} \quad \square$$

**Corollary 1** (1.13). *Let  $G$  be a group with a normal subgroup  $H$  of index  $n$ . Then  $G/H$  is isomorphic to a subgroup of  $S_n$ .*

*Proof.* Since  $H \trianglelefteq G$ , conjugation fixes  $H$ , so  $\text{core}(H) = \bigcap_{x \in G} xHx^{-1} = H$ . By Theorem 1.12,  $\ker \varphi = H$ , and by the First Isomorphism Theorem  $G/H \cong \text{im } \varphi \leq S_n$ .  $\square$

**Theorem 10** (1.24 — Orbit-Stabiliser Theorem). *Let  $G$  be a group acting on a set  $X$ . Then the set of all orbits in  $X$  under  $G$  is a partition of  $X$ . Furthermore, for each  $x \in X$  there is a bijection  $Gx \rightarrow G/G_x$ , and hence*

$$|Gx| = |G : G_x|.$$

*Therefore, if  $X$  is finite,*

$$|X| = \sum_{x \in C} |G : G_x|,$$

*where  $C \subseteq X$  contains exactly one element from each orbit.*

*Proof. (Partition.)* Define  $x \sim y$  if  $x = ay$  for some  $a \in G$ . This is an equivalence relation: reflexivity follows from  $e \cdot x = x$ , symmetry from  $x = ay \Rightarrow y = a^{-1}x$ , and transitivity from  $x = ay, y = bz \Rightarrow x = (ab)z$ . The equivalence class of  $x$  is exactly the orbit  $Gx$ , so the orbits partition  $X$ .

(*Bijection*  $Gx \rightarrow G/G_x$ .) Define

$$\varphi : Gx \rightarrow G/G_x, \quad ax \mapsto aG_x.$$

The key observation is the following chain of equivalences, for any  $a, b \in G$ :

$$ax = bx \iff a^{-1}bx = x \iff a^{-1}b \in G_x \iff aG_x = bG_x.$$

- *Well-defined:* If  $ax = bx$  (the same orbit element written using two different group elements), then  $aG_x = bG_x$ , so the coset  $\varphi$  assigns does not depend on the choice of representative.
- *Injective:* If  $\varphi(ax) = \varphi(bx)$ , i.e.  $aG_x = bG_x$ , then  $ax = bx$  by the chain above.
- *Surjective:* Take any coset  $aG_x \in G/G_x$ . Since  $a \in G$ , we have  $ax \in Gx$ , and  $\varphi(ax) = aG_x$ .

Hence  $\varphi$  is a bijection, giving  $|Gx| = |G/G_x| = |G : G_x|$ . Summing over one representative per orbit gives  $|X| = \sum_{x \in C} |G : G_x|$ .  $\square$

**Theorem 11** (1.25 — Class Equation). *Let  $G$  be a finite group. Then the conjugacy classes partition  $G$ , and*

$$|G| = \sum_{a \in C} |G : C_G(a)|,$$

where  $C \subseteq G$  contains exactly one element from each conjugacy class.

*Proof.* Let  $G$  act on itself by conjugation, i.e.  $a \cdot x = axa^{-1}$ . The orbit of  $x$  under this action is its conjugacy class  $\text{ccl}(x) = \{axa^{-1} \mid a \in G\}$ , and the stabiliser of  $x$  is  $G_x = \{a \in G \mid axa^{-1} = x\} = C_G(x)$ . Applying Theorem 1.24 directly gives the partition into orbits and  $|G| = \sum_{a \in C} |G : C_G(a)|$ .  $\square$

## Lecture 2

**Theorem 12** (2.1). *Let  $G$  be a finite group of order  $p^n$ , where  $p$  is prime and  $n \in \mathbb{N}$ . Then:*

- (i)  $G$  has a non-trivial centre  $Z(G)$ ;
- (ii)  $Z(G) \cap K$  is non-trivial for any non-trivial normal subgroup  $K$  of  $G$ ;
- (iii) if  $H$  is a proper subgroup of  $G$ , then  $H \subsetneq N_G(H)$ . In particular, if  $|H| = p^{n-1}$  then  $H \trianglelefteq G$ .

*Proof.* (i) Since  $x \in Z(G)$  iff  $\text{ccl}(x) = \{x\}$ , the class equation refines to

$$p^n = |Z(G)| + \sum_{x \in C} |G : C_G(x)|,$$

where  $C$  runs over conjugacy classes with more than one element. For each  $x \in C$ ,  $|C_G(x)|$  properly divides  $p^n$ , so  $p \nmid |G : C_G(x)|$ . Hence  $p$  divides the sum, and therefore  $p \mid |Z(G)|$ . Since  $e \in Z(G)$ , we have  $|Z(G)| \geq 1$ , so in fact  $|Z(G)| \geq p > 1$ .

(ii) Intersecting the class decomposition of  $G$  with  $K$  gives  $|K| = |Z(G) \cap K| + \sum_{x \in C} |\text{ccl}(x) \cap K|$ . Since  $K \trianglelefteq G$ , each conjugacy class is either disjoint from  $K$  or contained in  $K$ , so  $p \mid \sum |\text{ccl}(x) \cap K|$ . As  $p \mid |K|$ , we get  $p \mid |Z(G) \cap K|$ , and the same argument as (i) gives  $Z(G) \cap K$  non-trivial.

(iii) Let  $M$  be a maximal normal subgroup of  $G$  contained in  $H$ . Then  $G/M$  is a  $p$ -group, so by (i) it has a non-trivial centre  $L/M \trianglelefteq G/M$ . By the Correspondence Theorem,  $L \trianglelefteq G$ . Since  $M \subsetneq L$  strictly, maximality of  $M$  forces  $L \not\subseteq H$ .

It remains to show  $L \subseteq N_G(H)$ . To show  $\ell \in N_G(H)$  we need  $\ell^{-1}h\ell \in H$  for all  $h \in H$ . Take  $h \in H$  and  $\ell \in L$ . Since  $\ell M \in L/M = Z(G/M)$ , it commutes with every coset, so  $(\ell M)(hM) = (hM)(\ell M)$ , i.e.  $\ell hM = h\ell M$ . This gives  $\ell^{-1}h\ell \in hM$ . Since  $h \in H$  and  $M \subseteq H$ , we have  $hM \subseteq H$ , so  $\ell^{-1}h\ell \in H$ . Hence  $\ell \in N_G(H)$ , so  $L \subseteq N_G(H)$ .

Since  $L \subseteq N_G(H)$  but  $L \not\subseteq H$ , we have  $H \subsetneq N_G(H)$ . Finally, if  $|H| = p^{n-1}$  then  $N_G(H)$  has order strictly greater than  $p^{n-1}$ , and the only possibility is  $p^n = |G|$ , so  $N_G(H) = G$  and  $H \trianglelefteq G$ .  $\square$

**Theorem 13** (2.2 — Burnside's Lemma). *Let  $G$  be a finite group acting on a finite set  $X$ , and for  $g \in G$  let  $X_g := \{x \in X \mid gx = x\}$ . Then the number  $k$  of orbits in  $X$  under  $G$  is*

$$k = \frac{1}{|G|} \sum_{g \in G} |X_g|.$$

*Proof.* Let  $S := \{(g, x) \in G \times X \mid gx = x\}$ . Fixing  $g$  gives  $|S| = \sum_{x \in X} |X_g|$ ; fixing  $x$  gives  $|S| = \sum_{g \in G} |G_x|$ . Let  $C \subseteq X$  contain one element from each orbit. By the Orbit-Stabiliser Theorem,  $|G_x| = |G|/|Gx| = |G|/|Ga|$  for all  $x \in Ga$ . Hence

$$\frac{1}{|G|} \sum_{g \in G} |X_g| = \frac{|S|}{|G|} = \frac{1}{|G|} \sum_{x \in X} |G_x| = \frac{1}{|G|} \sum_{a \in C} \sum_{x \in Ga} \frac{|G|}{|Ga|} = \frac{1}{|G|} \sum_{a \in C} |G| = k. \quad \square$$

**Remark 1** (Pólya counting). *Burnside's Lemma counts distinct objects under symmetry. As an illustration: how many necklaces with  $p$  beads ( $p$  prime) can be made from  $n$  colours, where two necklaces are the same if one is a rotation of the other?*

*Let  $X$  be the set of all colourings, so  $|X| = n^p$ . The cyclic group  $G = \langle a \rangle$  of order  $p$  acts on  $X$  by rotation. For the identity,  $X_e = X$  so  $|X_e| = n^p$ . For any  $g \neq e$ : since  $|G|$  is prime,  $g$  generates  $G$ , so a necklace fixed by  $g$  must be fixed by all rotations, meaning all beads have the same colour. Hence  $|X_g| = n$ . Burnside then gives*

$$k = \frac{1}{p} (n^p + (p-1)n).$$

## Lecture 3

**Definition 2** ( $p$ -group, Sylow  $p$ -subgroup). *Let  $p$  be a prime. A group  $G$  is a  $p$ -group if every element of  $G$  has order a power of  $p$ . If  $G$  is finite and  $p^m \mid |G|$  but  $p^{m+1} \nmid |G|$ , then any subgroup of  $G$  of order  $p^m$  is called a Sylow  $p$ -subgroup of  $G$ .*

**Theorem 14** (3.5 — First Sylow Theorem). *Let  $G$  be a finite group and  $p$  a prime. If  $p^m \mid |G|$ , then  $G$  has a subgroup of order  $p^m$ .*

*Proof.* If  $m = 0$  the result is trivial. We proceed by induction on  $n = |G|$ , the case  $n = 1$  being trivial. Assume the result holds for all groups of order less than  $n$ .

*Case 1:*  $p \mid |Z(G)|$ . By the Fundamental Theorem of Finitely Generated Abelian Groups,  $Z(G)$  contains an element  $a$  of order  $p$ . Let  $C = \langle a \rangle$ . Since  $a \in Z(G)$ , we have  $C \trianglelefteq G$ , and  $|G/C| = n/p$ , which is divisible by  $p^{m-1}$ . By the induction hypothesis,  $G/C$  has a subgroup of order  $p^{m-1}$ , which by the Correspondence Theorem is of the form  $H/C$  for some  $H \leq G$ . Then  $|H| = |H/C| \cdot |C| = p^{m-1} \cdot p = p^m$ .

*Case 2:*  $p \nmid |Z(G)|$ . From the class equation  $n = |Z(G)| + \sum_a |G : C_G(a)|$ , since  $p \mid n$  but  $p \nmid |Z(G)|$ , there must exist some  $a \notin Z(G)$  with  $p \nmid |G : C_G(a)|$ . Since  $|G| = |G : C_G(a)| \cdot |C_G(a)|$  and  $p^m \mid |G|$  but  $p \nmid |G : C_G(a)|$ , it follows that  $p^m \mid |C_G(a)|$ . Since  $a \notin Z(G)$ , we have  $C_G(a) \neq G$ , so  $|C_G(a)| < n$ . By the induction hypothesis  $C_G(a)$  contains a subgroup of order  $p^m$ , which is also a subgroup of  $G$ .  $\square$

**Corollary 2** (3.6 — Cauchy's Theorem). *If  $p$  is a prime dividing  $|G|$ , then  $G$  has an element of order  $p$ .*

*Proof.* By the First Sylow Theorem,  $G$  has a subgroup  $H$  of order  $p$ . Every non-identity element of  $H$  has order  $p$ .  $\square$

**Corollary 3** (3.7). *A finite group  $G$  is a  $p$ -group if and only if  $|G|$  is a power of  $p$ .*

*Proof.* If  $|G|$  is a power of  $p$ , then by Lagrange every element order divides  $|G|$ , hence is a power of  $p$ . Conversely, if  $|G|$  is not a power of  $p$ , then some prime  $q \neq p$  divides  $|G|$ , and by Cauchy's Theorem  $G$  has an element of order  $q$ , contradicting that  $G$  is a  $p$ -group.  $\square$

**Theorem 15** (3.8 — Second Sylow Theorem). *Let  $G$  be a finite group and  $p$  a prime. All Sylow  $p$ -subgroups of  $G$  are conjugate.*

*Proof.* Let  $S, H \in \text{Syl}_p(G)$ . Let  $S$  act on the coset space  $G/H$  by left translation:  $s \cdot gH = sgH$ .

*Orbits have  $p$ -power size.* By the Orbit-Stabiliser Theorem, each orbit has size  $|S : S_{gH}|$ , which divides  $|S| = p^m$ , hence is a power of  $p$ .

*There must be a fixed point.* Since  $H$  is a Sylow  $p$ -subgroup,  $p \nmid |G|/|H| = |G/H|$ . Writing  $|G/H|$  as a sum of orbit sizes, each of which is a power of  $p$ , the only way this sum is not divisible by  $p$  is if at least one orbit has size  $p^0 = 1$ .

*A fixed point gives conjugacy.* If  $gH$  is a fixed point, then  $s \cdot gH = gH$  for all  $s \in S$ , so  $g^{-1}sg \in H$  for all  $s \in S$ . Hence  $g^{-1}Sg \subseteq H$ , and since  $|g^{-1}Sg| = |S| = |H|$ , we get  $g^{-1}Sg = H$ . So  $S$  and  $H$  are conjugate.  $\square$

**Corollary 4** (3.10). *A Sylow  $p$ -subgroup of  $G$  is unique if and only if it is normal in  $G$ .*

*Proof.* If  $P$  is the unique Sylow  $p$ -subgroup, then  $gPg^{-1} = P$  for all  $g \in G$  (since conjugates are also Sylow  $p$ -subgroups), so  $P \trianglelefteq G$ . Conversely, if  $P \trianglelefteq G$  then  $P$  is its own conjugate, so by the Second Sylow Theorem it is the only Sylow  $p$ -subgroup.  $\square$

**Theorem 16** (3.9 — Third Sylow Theorem). *Let  $G$  be a finite group and  $p$  a prime. The number  $n_p$  of Sylow  $p$ -subgroups satisfies  $n_p \mid |G|$  and  $n_p \equiv 1 \pmod{p}$ .*

*Proof.*  $n_p \mid |G|$ . By the Second Sylow Theorem, all Sylow  $p$ -subgroups are conjugate to a fixed  $K \in \text{Syl}_p(G)$ , so  $n_p = |\text{ccl}(K)| = |G : N_G(K)|$ , which divides  $|G|$ .

$n_p \equiv 1 \pmod{p}$ . Let  $P \in \text{Syl}_p(G)$ . Since conjugates of Sylow  $p$ -subgroups are again Sylow  $p$ -subgroups (by the Second Sylow Theorem),  $P$  acts on the set  $\text{Syl}_p(G)$  by conjugation:

$$x \cdot Q := xQx^{-1}, \quad x \in P, Q \in \text{Syl}_p(G).$$

This action partitions  $\text{Syl}_p(G)$  into orbits. Let  $F \subseteq \text{Syl}_p(G)$  denote the set of fixed points (those  $Q$  with  $xQx^{-1} = Q$  for all  $x \in P$ ), and let  $\mathcal{O}$  denote the collection of orbits of size greater than 1. Then:

$$n_p = |\text{Syl}_p(G)| = |F| + \sum_{O \in \mathcal{O}} |O|.$$

By the Orbit-Stabiliser Theorem each  $|O|$  divides  $|P| = p^m$ , so every  $O \in \mathcal{O}$  has size  $p^e$  for some  $e \geq 1$ , hence  $p \mid |O|$ . Therefore  $n_p \equiv |F| \pmod{p}$ .

It remains to show  $F = \{P\}$ , which gives  $n_p \equiv 1 \pmod{p}$ . Clearly  $P \in F$ . If  $Q \in F$ , then  $xQx^{-1} = Q$  for all  $x \in P$ , so  $P \leq N_G(Q)$ . By Corollary 3.10, since  $Q \trianglelefteq N_G(Q)$ ,  $Q$  is the unique Sylow  $p$ -subgroup of  $N_G(Q)$ . But  $P$  is also a Sylow  $p$ -subgroup of  $N_G(Q)$ , so  $P = Q$ . Hence  $F = \{P\}$ .  $\square$

## Lecture 4 — Modules: the Basics

**Definition 3** (4.1 — Left  $R$ -module). *Let  $R$  be a ring. An abelian group  $M$  is a left  $R$ -module if there is a map  $R \times M \rightarrow M$ ,  $(r, m) \mapsto rm$ , such that for all  $r, r_1, r_2 \in R$  and  $m, m_1, m_2 \in M$ :*

$$(i) \quad r(m_1 + m_2) = rm_1 + rm_2;$$

$$(ii) \quad (r_1 + r_2)m = r_1m + r_2m;$$

$$(iii) \quad (r_1r_2)m = r_1(r_2m);$$

$$(iv) \quad 1m = m \text{ if } 1 \in R.$$

**Lemma 1** (4.7). *Let  $M$  be an  $R$ -module. Then for  $a \in R$  and  $m \in M$ :*

$$(i) \quad 0m = 0;$$

$$(ii) \quad a0 = 0;$$

$$(iii) \quad (-a)m = -(am) = a(-m).$$

*Proof.* (i)  $0m = (0 + 0)m = 0m + 0m$ . Subtracting  $0m$  from both sides gives  $0 = 0m$ .

(ii)  $a0 = a(0 + 0) = a0 + a0$ . Subtracting  $a0$  from both sides gives  $0 = a0$ .

(iii) By (i) and axiom (ii):  $0 = 0m = (a + (-a))m = am + (-a)m$ , so  $(-a)m = -(am)$ . By (ii) and axiom (i):  $0 = a0 = a(m + (-m)) = am + a(-m)$ , so  $a(-m) = -(am)$ .  $\square$

**Definition 4** (4.8 — Submodule). *A subset  $N$  of an  $R$ -module  $M$  is an  $R$ -submodule of  $M$  if:*

$$(i) \quad N \neq \emptyset;$$

$$(ii) \quad a - b \in N \text{ for all } a, b \in N;$$

$$(iii) \quad ra \in N \text{ for all } a \in N, r \in R.$$

**Theorem 17** (4.12). *Let  $(N_i)_{i \in I}$  be a family of  $R$ -submodules of an  $R$ -module  $M$ . Then  $\bigcap_{i \in I} N_i$  is also an  $R$ -submodule of  $M$ .*

*Proof.* Since  $0 \in N_i$  for all  $i$ , we have  $0 \in \bigcap N_i$ , so it is non-empty. Let  $x, y \in \bigcap N_i$  and  $r \in R$ . For each  $i$ , since  $N_i$  is a submodule,  $x - y \in N_i$  and  $rx \in N_i$ . Hence  $x - y$  and  $rx$  lie in  $\bigcap N_i$ .  $\square$

**Definition 5** (4.13 — Generated submodule, cyclic module). *Let  $S \subseteq M$ . The family  $\mathcal{A} = \{N \mid N \text{ is an } R\text{-submodule of } M \text{ containing } S\}$  is non-empty (since  $M \in \mathcal{A}$ ), so by Theorem 4.12 the intersection  $\bigcap_{N \in \mathcal{A}} N$  is an  $R$ -submodule. It is the smallest submodule of  $M$  containing  $S$ , called the submodule generated by  $S$  and denoted  $(S)$ . If  $S = \{x_1, \dots, x_n\}$  we write  $(x_1, \dots, x_n)$ . An  $R$ -module  $M$  is cyclic if  $M = (x)$  for some  $x \in M$ .*

**Theorem 18** (4.15). *Let  $R$  be a ring with unity and  $x_1, \dots, x_n \in M$ . Then*

$$(x_1, \dots, x_n) = \sum_{i=1}^n Rx_i := \{r_1x_1 + \dots + r_nx_n \mid r_i \in R\}.$$

*Proof.*  $(\supseteq)$  The submodule  $(x_1, \dots, x_n)$  contains each  $x_i$ . By closure under scalar multiplication it contains  $rx_i$  for all  $r \in R$ , and by closure under addition it contains every sum  $r_1x_1 + \dots + r_nx_n$ . Hence  $\sum Rx_i \subseteq (x_1, \dots, x_n)$ .

$(\subseteq)$  We show  $\sum Rx_i$  is a submodule containing each  $x_i$ ; then  $(x_1, \dots, x_n) \subseteq \sum Rx_i$  follows from minimality of  $(x_1, \dots, x_n)$ .

Each  $x_i \in \sum Rx_i$  since  $x_i = 1 \cdot x_i$  (using unity). For the submodule conditions:  $0 = \sum 0 \cdot x_i \in \sum Rx_i$ , so it is non-empty. For  $\sum r_ix_i, \sum s_ix_i \in \sum Rx_i$  and  $r \in R$ ,

$$\sum r_ix_i - \sum s_ix_i = \sum (r_i - s_i)x_i \in \sum Rx_i, \quad r \sum r_ix_i = \sum (rr_i)x_i \in \sum Rx_i. \quad \square$$

**Definition 6** (4.18 — Sum of submodules). *Let  $N_1, \dots, N_k$  be  $R$ -submodules of  $M$ . Their sum  $\sum_{i=1}^k N_i$  is the submodule of  $M$  generated by  $\bigcup_{i=1}^k N_i$ , i.e. the smallest submodule of  $M$  containing each  $N_i$ .*

## Lecture 5 — Modules: continuation

**Theorem 19** (4.19). *Let  $N_1, \dots, N_k$  be  $R$ -submodules of an  $R$ -module  $M$ . Then*

$$\sum_{i=1}^k N_i = \{x_1 + \dots + x_k \mid x_i \in N_i\}.$$

*Proof.* Let

$$A = \{x_1 + \dots + x_k \mid x_i \in N_i\}.$$

It is straightforward to check that  $A$  is a submodule of  $M$ , and it contains every  $N_i$ . Hence, by minimality,

$$\sum_{i=1}^k N_i \subseteq A.$$

Conversely,  $\sum_{i=1}^k N_i$  contains every  $N_i$  and is closed under addition, so it contains every element of  $A$ . Therefore

$$A \subseteq \sum_{i=1}^k N_i.$$

□

**Definition 7** (4.21 — Direct sum). *The sum  $N_1 + \cdots + N_k$  is direct if every element has a unique representation*

$$x = x_1 + \cdots + x_k, \quad x_i \in N_i.$$

In this case we write

$$N_1 \oplus \cdots \oplus N_k.$$

**Theorem 20** (4.22). *Let  $N_1, \dots, N_k$  be submodules of an  $R$ -module  $M$ . The following are equivalent:*

- (i)  $\sum_{i=1}^k N_i$  is direct;
- (ii) if  $\sum_{i=1}^k x_i = 0$ , where  $x_i \in N_i$ , then every  $x_i = 0$ ;
- (iii) for every  $i$ ,

$$N_i \cap \sum_{j \neq i} N_j = \{0\}.$$

*Proof.* (i)  $\Rightarrow$  (ii): Zero already has the representation  $0 + \cdots + 0$ , so uniqueness implies that every  $x_i = 0$ .

(ii)  $\Rightarrow$  (iii): If

$$x \in N_i \cap \sum_{j \neq i} N_j,$$

then  $x = \sum_{j \neq i} x_j$  for some  $x_j \in N_j$ . Hence

$$x - \sum_{j \neq i} x_j = 0,$$

so (ii) gives  $x = 0$ .

(iii)  $\Rightarrow$  (i): Suppose

$$\sum_{i=1}^k x_i = \sum_{i=1}^k y_i, \quad x_i, y_i \in N_i.$$

For every  $i$ ,

$$x_i - y_i = - \sum_{j \neq i} (x_j - y_j) \in N_i \cap \sum_{j \neq i} N_j = \{0\}.$$

Thus  $x_i = y_i$  for every  $i$ , so the representation is unique.

□

**Definition 8** (4.23 — *R*-homomorphism). Let  $M$  and  $N$  be  $R$ -modules. A map  $f: M \rightarrow N$  is an  $R$ -homomorphism if, for all  $x, y \in M$  and  $r \in R$ ,

$$f(x + y) = f(x) + f(y), \quad f(rx) = rf(x).$$

The set of all  $R$ -homomorphisms  $M \rightarrow N$  is denoted

$$\text{Hom}_R(M, N).$$

An  $R$ -homomorphism  $M \rightarrow M$  is called an endomorphism, and

$$\text{End}_R(M) := \text{Hom}_R(M, M).$$

**Theorem 21** (4.30). Let  $M = \bigoplus_{i=1}^k M_i$  be a direct sum of  $R$ -modules. Then

$$\text{Hom}_R(M, M) \cong (\text{Hom}_R(M_j, M_i))_{1 \leq i, j \leq k}$$

as rings, where the right-hand side is a ring  $T$  of  $k \times k$  matrices under the usual matrix addition and multiplication, with  $(i, j)$  entry in  $\text{Hom}_R(M_j, M_i)$ .

*Proof.* Let  $\lambda_j: M_j \rightarrow M$  be the natural inclusion and  $\pi_i: M \rightarrow M_i$  the natural projection. Two key identities hold:

$$\sum_{\ell=1}^k \lambda_\ell \pi_\ell = \text{id}_M, \quad \pi_p \lambda_q = \delta_{pq} \text{id}_{M_p}.$$

The first says every  $x \in M$  equals the sum of its components; the second says projecting the  $q$ -th inclusion gives zero unless  $p = q$ .

Define  $\sigma: \text{Hom}_R(M, M) \rightarrow T$  by  $\sigma(\phi) = (\pi_i \phi \lambda_j)_{ij}$ .

*Ring homomorphism.* Additivity is immediate. For multiplication, using the first identity:

$$(\sigma(\phi)\sigma(\psi))_{ij} = \sum_{\ell} \pi_i \phi \lambda_\ell \pi_\ell \psi \lambda_j = \pi_i \phi \left( \sum_{\ell} \lambda_\ell \pi_\ell \right) \psi \lambda_j = \pi_i \phi \text{id}_M \psi \lambda_j = (\sigma(\phi\psi))_{ij}.$$

*Injective.* Suppose  $\sigma(\phi) = 0$ , so  $\pi_i \phi \lambda_j = 0$  for all  $i, j$ . Applying  $\lambda_i$  on the left and summing over  $i$  using the first identity:

$$\phi \lambda_j = \text{id}_M \cdot \phi \lambda_j = \left( \sum_i \lambda_i \pi_i \right) \phi \lambda_j = \sum_i \lambda_i (\pi_i \phi \lambda_j) = 0.$$

Now applying  $\pi_j$  on the right and summing over  $j$ :

$$\phi = \phi \cdot \text{id}_M = \phi \left( \sum_j \lambda_j \pi_j \right) = \sum_j (\phi \lambda_j) \pi_j = 0.$$

*Surjective.* Given  $f = (f_{ij}) \in T$ , set  $\phi = \sum_{i,j} \lambda_i f_{ij} \pi_j$ . Using the second identity:

$$(\sigma(\phi))_{st} = \pi_s \phi \lambda_t = \sum_{i,j} \pi_s \lambda_i f_{ij} \pi_j \lambda_t = \sum_{i,j} \delta_{si} f_{ij} \delta_{jt} = f_{st}.$$

Hence  $\sigma(\phi) = f$ , so  $\sigma$  is surjective. □

**Definition 9** (Quotient module). Let  $N$  be an  $R$ -submodule of an  $R$ -module  $M$ . The set of cosets  $M/N = \{a + N \mid a \in M\}$  becomes an  $R$ -module under

$$(a + N) + (b + N) = (a + b) + N, \quad r(a + N) = ra + N.$$

It is called the quotient module of  $M$  by  $N$ .

**Theorem 22** (4.31 — Correspondence Theorem for Modules). Let  $N$  be an  $R$ -submodule of an  $R$ -module  $M$ . Every  $R$ -submodule of  $M/N$  is of the form  $U/N$ , where  $U$  is an  $R$ -submodule of  $M$  containing  $N$ .

*Proof.* The canonical projection  $f : M \rightarrow M/N$ ,  $f(x) = x + N$ , is an  $R$ -homomorphism:

$$f(x+y) = (x+y)+N = (x+N)+(y+N) = f(x)+f(y), \quad f(rx) = rx+N = r(x+N) = rf(x).$$

Let  $X$  be an  $R$ -submodule of  $M/N$ . The structure of the proof is: define  $U$  as the preimage of  $X$  under  $f$ , show  $U$  is a submodule of  $M$  containing  $N$ , and conclude  $X = U/N$ .

Set  $U = f^{-1}(X) = \{x \in M \mid f(x) \in X\}$ .

$N \subseteq U$ . For any  $x \in N$ ,  $f(x) = x + N = 0 + N = 0_{M/N} \in X$  (since  $X$  is a submodule, it contains 0). Hence  $x \in U$ .

$U$  is an  $R$ -submodule. Since  $0 \in N \subseteq U$ ,  $U$  is non-empty. For  $x, y \in U$  and  $r \in R$ , since  $f$  is an  $R$ -homomorphism:

$$f(x - y) = f(x) - f(y) \in X, \quad f(rx) = rf(x) \in X,$$

so  $x - y \in U$  and  $rx \in U$ .

$X = U/N$ . By definition of  $U$ ,  $f(U) \subseteq X$ . For the reverse inclusion, take any  $x \in X$ ; since  $f$  is surjective there exists  $y \in M$  with  $f(y) = x$ , so  $y \in U$  and  $x \in f(U)$ . Hence  $X = f(U) = U/N$ .  $\square$

**Theorem 23** (4.32 — First Isomorphism Theorem for Modules). Let  $f : M \rightarrow N$  be an  $R$ -homomorphism. Then  $M/\ker f \cong f(M)$ .

*Proof.* Define  $g : M/\ker f \rightarrow f(M)$  by  $g(x + \ker f) = f(x)$ . We know from group theory that  $g$  is an isomorphism of abelian groups (well-defined and injective because  $x + \ker f = y + \ker f \iff x - y \in \ker f \iff f(x) = f(y)$ ; surjective onto  $f(M)$  by definition). It remains to check that  $g$  respects the  $R$ -action:

$$g(r(x + \ker f)) = g(rx + \ker f) = f(rx) = rf(x) = rg(x + \ker f). \quad \square$$

**Theorem 24** (Second Isomorphism Theorem for Modules). Let  $U$  and  $V$  be  $R$ -submodules of an  $R$ -module  $M$ . Then  $(U + V)/V \cong U/(U \cap V)$ .

*Proof.* Define  $f : U \rightarrow (U + V)/V$  by  $f(u) = u + V$ .

$f$  is an  $R$ -homomorphism. For  $u_1, u_2 \in U$  and  $r \in R$ :

$$f(u_1 + u_2) = (u_1 + u_2) + V = (u_1 + V) + (u_2 + V) = f(u_1) + f(u_2),$$

$$f(ru) = ru + V = r(u + V) = rf(u).$$

$f$  is surjective. Any element of  $(U + V)/V$  has the form  $(u + v) + V = u + V = f(u)$  for some  $u \in U$ ,  $v \in V$  (since  $v \in V$  means  $v + V = V$ ).

$\ker f = U \cap V$ . We have  $f(u) = 0$  iff  $u + V = V$  iff  $u \in V$ , so  $\ker f = \{u \in U \mid u \in V\} = U \cap V$ .

By the First Isomorphism Theorem,  $U/(U \cap V) \cong (U + V)/V$ .  $\square$

**Theorem 25** (Third Isomorphism Theorem for Modules). *Let  $U \subseteq V$  be  $R$ -submodules of an  $R$ -module  $M$ . Then  $(M/U)/(V/U) \cong M/V$ .*

*Proof.* Define  $f : M/U \rightarrow M/V$  by  $f(x+U) = x+V$ .

$f$  is well-defined. If  $x+U = y+U$  then  $x-y \in U \subseteq V$ , so  $x+V = y+V$ .

$f$  is an  $R$ -homomorphism. For  $x+U, y+U \in M/U$  and  $r \in R$ :

$$f((x+U)+(y+U)) = f((x+y)+U) = (x+y)+V = (x+V)+(y+V) = f(x+U)+f(y+U),$$

$$f(r(x+U)) = f(rx+U) = rx+V = r(x+V) = rf(x+U).$$

$f$  is surjective. For any  $x+V \in M/V$ ,  $f(x+U) = x+V$ .

$\ker f = V/U$ . We have  $f(x+U) = 0$  iff  $x+V = V$  iff  $x \in V$ , so  $\ker f = \{x+U \mid x \in V\} = V/U$ .

By the First Isomorphism Theorem,  $(M/U)/(V/U) \cong M/V$ .  $\square$

**Theorem 26** (4.33). *Let  $A \leq M$  and  $B \leq N$  be  $R$ -submodules. Then  $(M \times N)/(A \times B) \cong M/A \times N/B$ .*

*Proof.* Define  $f : M \times N \rightarrow M/A \times N/B$  by  $f(m, n) = (m+A, n+B)$ . This is a surjective  $R$ -homomorphism, and  $\ker f = \{(m, n) \mid m \in A, n \in B\} = A \times B$ . By the First Isomorphism Theorem,  $(M \times N)/(A \times B) \cong M/A \times N/B$ .  $\square$

## Lecture 6 — Simple, Completely Reducible, and Free Modules

**Definition 10** (5.1 — Simple module). *Let  $R$  be a ring with unity. An  $R$ -module  $M$  is called simple (or irreducible) if  $M \neq 0$  and the only  $R$ -submodules of  $M$  are  $0$  and  $M$ .*

**Theorem 27** (5.5 — Characterisation of Simple Modules). *Let  $R$  be a ring with unity and  $M$  an  $R$ -module. The following are equivalent:*

- (i)  $M$  is simple;
- (ii)  $M \neq 0$ , and  $M$  is generated by any  $0 \neq x \in M$ ;
- (iii)  $M \cong R/I$ , where  $I$  is a maximal left ideal of  $R$ .

*Proof.* (i)  $\Rightarrow$  (ii): Let  $0 \neq x \in M$ . Then  $Rx$  is a non-zero submodule of  $M$ , so  $Rx = M$  by simplicity.

(ii)  $\Rightarrow$  (iii): Take  $0 \neq x \in M$  and define  $f : R \rightarrow M$  by  $f(r) = rx$ . This is a surjective  $R$ -homomorphism, so letting  $I = \ker f$ , the First Isomorphism Theorem gives  $R/I \cong M$ . Note that (ii) implies  $M$  is simple: if  $0 \neq N \leq M$  and  $0 \neq x \in N$ , then  $M = Rx \subseteq N$ . Hence  $R/I \cong M$  is simple. By the Correspondence Theorem (4.31), any left ideal  $J$  with  $I \subseteq J \subseteq R$  corresponds to a submodule  $J/I$  of  $R/I$ . Simplicity forces  $J/I = 0$  or  $J/I = R/I$ , i.e.  $J = I$  or  $J = R$ , so  $I$  is maximal.

(iii)  $\Rightarrow$  (i): Let  $0 \neq N \leq M \cong R/I$ . By the Correspondence Theorem (4.31),  $N \cong J/I$  for some left ideal  $J$  with  $I \subsetneq J \subseteq R$ . Since  $I$  is maximal,  $J = R$ , so  $N = M$ .  $\square$

**Theorem 28** (5.6 — Schur's Lemma). *Let  $M$  be a simple  $R$ -module. Then  $\text{Hom}_R(M, M)$  is a division ring.*

*Proof.* Let  $0 \neq f \in \text{Hom}_R(M, M)$ . We show  $f$  is invertible and  $f^{-1} \in \text{Hom}_R(M, M)$ .

$f$  is bijective. Since  $\ker f$  and  $\text{im } f$  are submodules of  $M$ , simplicity forces each to be 0 or  $M$ . As  $f \neq 0$ , we have  $\ker f \neq M$  and  $\text{im } f \neq 0$ , so  $\ker f = 0$  (injective) and  $\text{im } f = M$  (surjective).

$f^{-1}$  is an  $R$ -homomorphism. We cannot push  $f^{-1}$  through operations directly, but we can push  $f$  through them. The strategy is: to prove  $f^{-1}(u) = v$ , it suffices to show  $f(u) = f(v)$ , since  $f$  is injective. For  $x, y \in M$  and  $r \in R$ :

$$\begin{aligned} f(f^{-1}(x + y)) &= x + y = f(f^{-1}(x)) + f(f^{-1}(y)) = f(f^{-1}(x) + f^{-1}(y)), \\ f(f^{-1}(rx)) &= rx = rf(f^{-1}(x)) = f(rf^{-1}(x)), \end{aligned}$$

where we used that  $f$  is itself an  $R$ -homomorphism in the last step of each line. Injectivity of  $f$  gives  $f^{-1}(x + y) = f^{-1}(x) + f^{-1}(y)$  and  $f^{-1}(rx) = rf^{-1}(x)$ .  $\square$

**Definition 11** (5.7 — Completely reducible module). Let  $R$  be a ring with unity. An  $R$ -module  $M$  is called completely reducible (or semisimple) if  $M = \sum_{\alpha \in \Lambda} M_\alpha$ , where each  $M_\alpha$  is a simple  $R$ -submodule.

**Theorem 29** (5.9). Let  $M = \sum_{\alpha \in \Lambda} M_\alpha$  be a sum of simple  $R$ -submodules, and let  $K$  be a submodule of  $M$ . Then there exists  $\Lambda' \subseteq \Lambda$  such that  $M = K \oplus (\bigoplus_{\alpha \in \Lambda'} M_\alpha)$ .

*Proof.* Let  $S$  be the family of subsets  $A \subseteq \Lambda$  satisfying two conditions:  $\bigoplus_{\alpha \in A} M_\alpha$  is a direct sum, and  $K \cap \bigoplus_{\alpha \in A} M_\alpha = 0$ . Together these mean  $K \oplus \bigoplus_{\alpha \in A} M_\alpha$  is a genuine direct sum. Since  $\emptyset$  satisfies both conditions trivially,  $S \neq \emptyset$ . Partially order  $S$  by inclusion; every chain  $(A_i)$  in  $S$  has upper bound  $\bigcup A_i$  in  $S$ . By Zorn's Lemma,  $S$  has a maximal element  $A$ .

Set  $N = K \oplus \bigoplus_{\alpha \in A} M_\alpha$ . We claim  $N = M$ . Fix any  $\beta \in \Lambda$ . Since  $M_\beta$  is simple,  $M_\beta \cap N$  is either 0 or  $M_\beta$ . If  $M_\beta \cap N = 0$ , then  $M_\beta$  intersects neither  $K$  nor  $\bigoplus_{\alpha \in A} M_\alpha$ , so  $A \cup \{\beta\}$  would satisfy both conditions in  $S$ , contradicting maximality of  $A$ . Hence  $M_\beta \cap N = M_\beta$ , i.e.  $M_\beta \subseteq N$ , for every  $\beta \in \Lambda$ . Since  $M = \sum_{\beta} M_\beta$ , we conclude  $N = M$ .  $\square$

**Corollary 5** (5.10). If  $M = \sum_{\alpha \in \Lambda} M_\alpha$  is a sum of simple  $R$ -submodules, then there exists  $\Lambda' \subseteq \Lambda$  such that  $M = \bigoplus_{\alpha \in \Lambda'} M_\alpha$ .

*Proof.* Apply Theorem 5.9 with  $K = \{0\}$ .  $\square$

**Definition 12** (5.11 — Linear independence and basis). A sequence  $x_1, \dots, x_n$  in an  $R$ -module  $M$  is linearly independent if  $\sum_{i=1}^n a_i x_i = 0$  implies  $a_1 = \dots = a_n = 0$  for any  $a_i \in R$ . A subset  $S \subseteq M$  is linearly independent if every finite sequence of distinct elements of  $S$  is linearly independent.

A subset  $B \subseteq M$  is a basis of  $M$  if  $B$  generates  $M$  and  $B$  is linearly independent. An  $R$ -module that has a basis is called a free module.

## Lecture 7 — Free Modules and Chain Conditions

**Theorem 30** (5.19). Let  $M$  be a free  $R$ -module with basis  $\{f_1, \dots, f_n\}$ . Then  $M \cong R^n$  as  $R$ -modules.

*Proof.* Define  $\varphi : M \rightarrow R^n$  by  $\varphi(\sum_{i=1}^n r_i f_i) = (r_1, \dots, r_n)$ , i.e. read off the coordinates in the basis. Since every element of  $M$  has a unique representation as  $\sum r_i f_i$  (by linear independence of the  $f_i$ ),  $\varphi$  is well-defined. It is clearly an  $R$ -homomorphism, injective (if  $(r_1, \dots, r_n) = 0$  then each  $r_i = 0$ , so  $m = 0$ ), and surjective (every  $(r_1, \dots, r_n) \in R^n$  is the image of  $\sum r_i f_i$ ).  $\square$

**Theorem 31** (5.20). *Let  $M$  be a finitely generated free  $R$ -module. Then every basis of  $M$  is finite.*

*Proof.* Let  $\{e_i\}_{i \in \Lambda}$  be a basis and  $\{x_1, \dots, x_n\}$  a finite generating set. Since elements of a module are finite sums by definition, each  $x_j$  involves only finitely many nonzero coefficients, so is supported on a finite subset  $S_j \subseteq \Lambda$ . Every basis element must appear in the expression of some generator (otherwise it could not be reached), so  $\Lambda \subseteq S_1 \cup \dots \cup S_n$ , a finite union of finite sets. Hence  $\Lambda$  is finite.  $\square$

**Theorem 32** (5.21). *Let  $M$  be a finitely generated free module over a commutative ring  $R$ . Then all bases of  $M$  have the same number of elements.*

*Proof.* Suppose  $M$  has two bases of sizes  $m$  and  $n$  respectively. By Theorem 5.19,  $M \cong R^m$  and  $M \cong R^n$ , so  $R^m \cong R^n$ . It suffices to show  $m = n$ .

Assume for contradiction  $m < n$ . Let  $\phi : R^m \rightarrow R^n$  be an  $R$ -isomorphism with inverse  $\psi = \phi^{-1}$ . Let  $(e_1, \dots, e_m)$  and  $(f_1, \dots, f_n)$  be the standard bases of  $R^m$  and  $R^n$ . Since  $\phi$  is an  $R$ -homomorphism, it is determined by where it sends each  $e_i$ , and each image lands in  $R^n$ , so we can write

$$\phi(e_i) = \sum_{j=1}^n a_{ji} f_j, \quad \psi(f_j) = \sum_{k=1}^m b_{kj} e_k.$$

Let  $A = (a_{ji})$  be the resulting  $n \times m$  matrix and  $B = (b_{kj})$  the  $m \times n$  matrix. Since  $\psi\phi = \text{id}_{R^m}$  and  $\phi\psi = \text{id}_{R^n}$ , substituting the expressions above and using linear independence of the bases gives  $BA = I_m$  and  $AB = I_n$ .

Pad these to  $n \times n$  matrices by setting  $A' = \begin{pmatrix} A & 0 \end{pmatrix}$  and  $B' = \begin{pmatrix} B \\ 0 \end{pmatrix}$ . Then  $A'B' = I_n$ , so  $\det(A'B') = 1$ . But  $B'A' = \begin{pmatrix} I_m & 0 \\ 0 & 0 \end{pmatrix}$ , so  $\det(B'A') = 0$ . Since  $R$  is commutative, determinants multiply, giving

$$1 = \det(A'B') = \det(A') \det(B') = \det(B'A') = 0,$$

a contradiction. By symmetry  $n < m$  also fails, so  $m = n$ .  $\square$